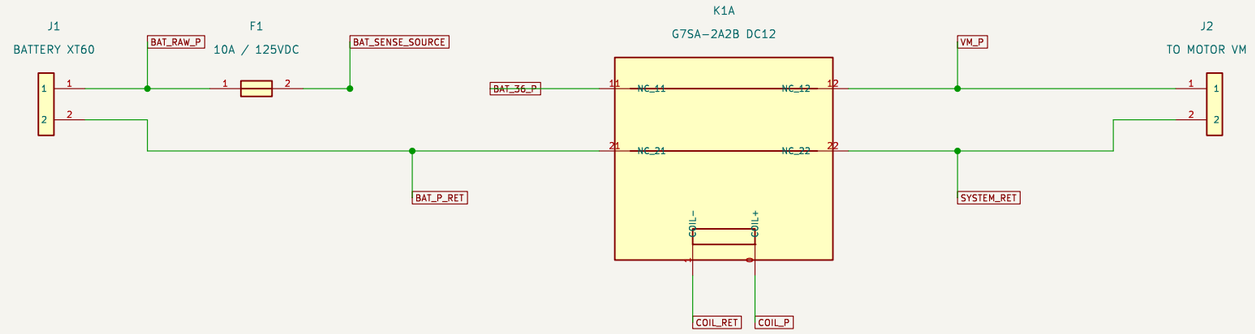
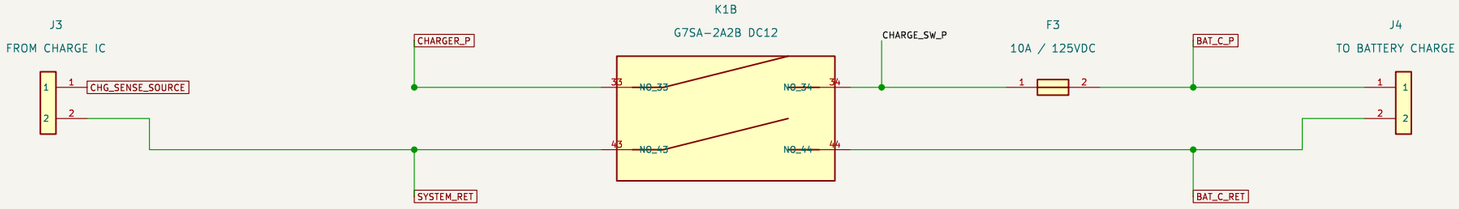
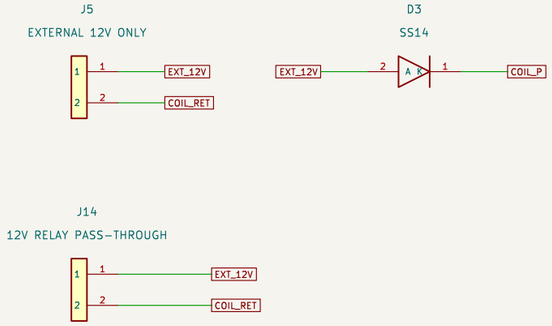
01 / DRIVE + CHARGE TRANSFER



12 V ABSENT: K1 NC drive contacts closed / NO charge contacts open.
12 V PRESENT: K1 NC drive contacts open / NO charge contacts closed.
K1A and K1B are ONE physical, forcibly guided 2NC + 2NO relay.

External controller: disable drive before transfer.
Assert 12 V, wait 1 s, then AC ON.
AC OFF, wait 1 s, then remove 12 V.
Near-zero-current transfer; no loaded 36–54.6 V switching approval.
No precharge or AC switching on this PCB.





One 12 V / 30 mA relay coil, separate from battery and SYSTEM_RET.
Coil terminal 0 = positive; terminal 1 = negative.
3 A maximum HEADER rating; actual jumper leads may be lower.
Retained 10 A fuses do not protect jumper wiring from overload.

J14 duplicates J5 BEFORE D3: 1 = external +12 V, 2 = coil return.
To motor J11.1 Relay+ / J11.2 Relay−. Use ONE switched source.
Shared source: approximately 0.20 A with five 12 V G5Q-1 coils; allow >=0.3 A.
No connection to battery or SYSTEM_RET. Motor relay suppression remains external.

Isolated Supply

File: Isolated_Supply.kicad_sch

Low Voltage IO

File: Low_Voltage_IO.kicad_sch

Charge Mode

File: Charge_Mode.kicad_sch

Buck 36V

File: Buck_36V.kicad_sch

VM Transient Clamp

File: VM_Transient_Clamp.kicad_sch

SOC Sensing

File: SOC_Sensing.kicad_sch



84 x 128 mm | 4 layers / 1 oz | 36 V nominal | isolated low-voltage system

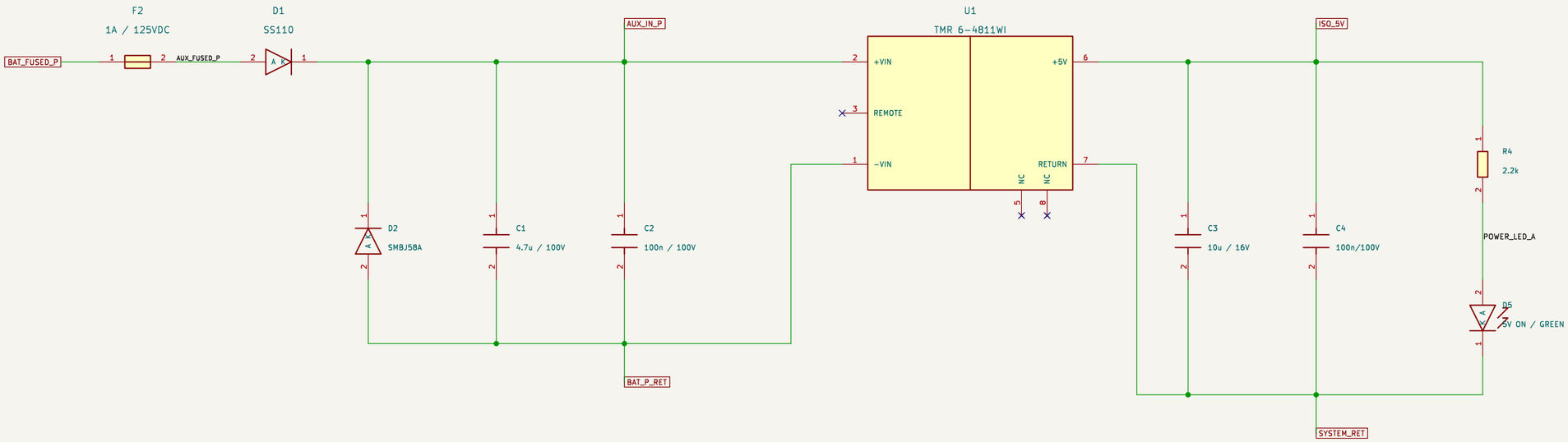
NIU intermediate board

Sheet: /
File: NIU_Intermediate_RevO.kicad_sch

Title: Battery / onboard transfer relays

Size: A3	Date: 2026-10-08	**Rev: O PROTOTYPE**
KiCad E.D.A. 9.0.7		Id: 1/7

02 / ALWAYS-ON ISOLATED SUPPLY



Battery power is taken BEFORE the drive relay: supply stays on in charge mode.
U1 remote pin is deliberately open. Pins 5 and 8 are NC.
Functional isolation for an isolated low-voltage system; no mains safety barrier claim.

Initial total 5 V load budget: 500 mA across MCU + comm + throttle.
Module capability: 1.2 A; validate actual load, startup and temperature before increasing.
F1/F3 remain 10 A. F2 independently protects the small auxiliary branch.

D1 protects ONLY the auxiliary converter against reversed battery polarity.
The main VM path now includes a buck and reverse-blocking diode. Check XT60 polarity.
D2 is a pulse clamp; sustained input overvoltage is outside the design envelope.

H1
M3 / NYLON

H2
M3 / NYLON

H3
M3 / NYLON

H4
M3 / NYLON

AUX_IN_P

SYSTEM_RET

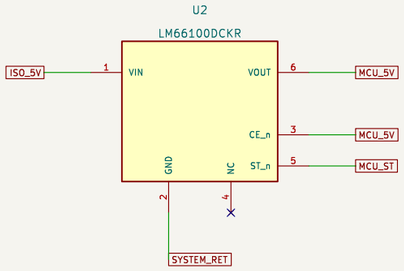
84 x 128 mm | 4 layers / 1 oz | 36 V nominal | isolated low-voltage system
NIU intermediate board

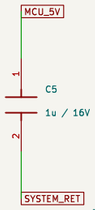
Sheet: /Isolated Supply/
File: Isolated_Supply.kicad_sch

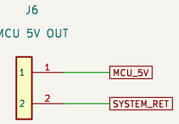
Title: Always-on isolated 5 V supply

Size: A3	Date: 2026-10-08	**Rev: 0 PROTOTYPE**
KiCad E.D.A. 9.0.7		Id: 2/7

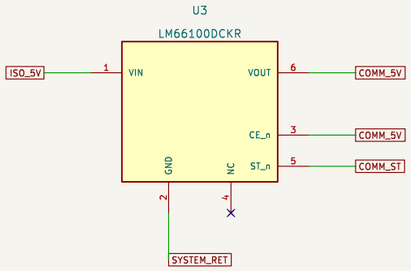
03 / LOW-VOLTAGE INTERFACES

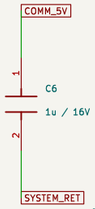


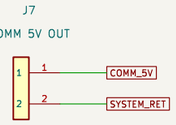




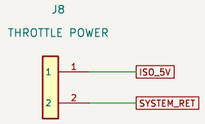


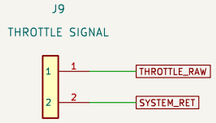


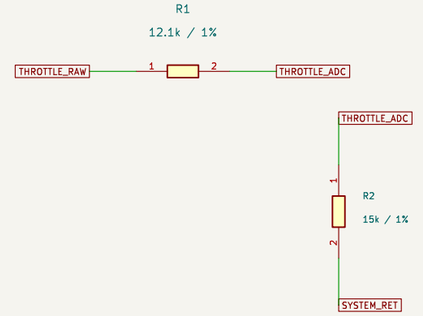


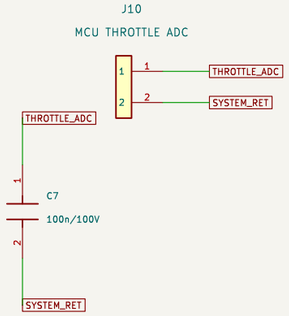












U2/U3 prevent external 5 V supplies backfeeding the shared converter rail.
They do not isolate the two downstream grounds from one another.
All SYSTEM_RET pins connect to the motor-board MCU ground.

External throttle: ElectricScooterParts THR-38.
Red = J8.1 (+5 V); black = J8.2 (return); white = J9.1 (signal).
J9.2 is the same return; use a suitable adapter harness.
Nominal 1-4 V signal becomes 0.554-2.214 V at J10.1.
J10.2 = return. Connect to motor-board PA6 ADC + ground.
Firmware must calibrate throttle and require idle-to-arm.

Throttle is powered entirely from the isolated MCU-side supply.
No separate analog isolator is required within that same domain.
MCU and comm use two separate 5 V/return output headers.
5 V + return and signal + return split the 3-wire throttle over two headers.

84 x 128 mm | 4 layers / 1 oz | 36 V nominal | isolated low-voltage system

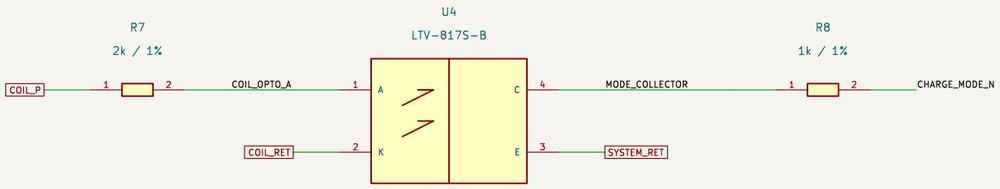
NIU intermediate board

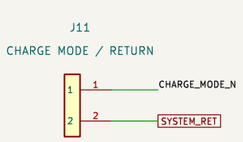
Sheet: /Low Voltage IO/
File: Low_Voltage_IO.kicad_sch

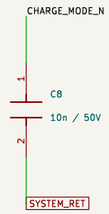
Title: MCU, communications and throttle interfaces

Size: A3	Date: 2026-10-08	**Rev: 0 PROTOTYPE**
KiCad E.D.A. 9.0.7		Id: 3/7

04 / ISOLATED CHARGE-MODE STATUS







12 V ABSENT: optocoupler off; MCU pull-up reads HIGH = drive mode.
12 V PRESENT: optocoupler on; input LOW = charge mode.
MCU, comm and throttle 5 V stay powered in both modes.

J11.1 -> motor board J16.1 -> U4 pin 2 / PC13.
J11.2 -> motor board ground, e.g. J4.5 or J9.5.
Configure PC13 as a digital INPUT WITH PULL-UP to its own 3.3 V.
Do not configure push-pull output; no 5 V pull-up is fitted on this board.

Firmware must keep gate-driver/PWM outputs disabled from reset,
read mode before arming, and inhibit drive immediately in charge mode.
Re-arm only after 12 V is absent, settling delay and idle throttle.
This signal senses the 12 V command, not relay contact position.
An unplugged signal wire reads DRIVE: this is not a safety interlock.

84 x 128 mm | 4 layers / 1 oz | 36 V nominal | isolated low-voltage system

NIU intermediate board

Sheet: /Charge Mode/
File: Charge_Mode.kicad_sch

Title: Isolated charge-mode input for PC13

Size: A3	Date: 2026-10-08	**Rev: 0 PROTOTYPE**
KiCad E.D.A. 9.0.7		Id: 4/7

05 / BATTERY-SIDE BUCK + REVERSE BLOCK

R9
442k
BAT_BUCK_RAW
BAT_FB_SERIES

R14
383k
BAT_FUSED_P
BAT_EN

L1
33uH / SRP1265A
BAT_SW
BAT_ISNS

U5
LM70860RRXR
1 VIN4
2 VIN5
3 VIN6
4 CBOOT
5 SW4
6 BIAS
7 PG
8 PFM
9 EN
10 NC
11 ISNS+
12 VOUT
13 CONFIG
14 RT
15 EXTCOMP
16 FB
17 AGND
18 VDDA
19 VCC
20 SW1
21 SW2
22 SW3
23 PGND1
24 PGND2
25 PGND3
26 PGND4
27 VIN1
28 VIN2
29 VIN3
30 PGND

R10
10k
BAT_FB
BAT_P_RET

R15
10k
BAT_EN
BAT_P_RET

R16
6m / 1%
BAT_ISNS
BAT_BUCK_RAW

R11
49.9k
BAT_RT
BAT_P_RET

R17
1R
BAT_BOOT_R
BAT_BOOT

D7
STPST10H100SB
BAT_BUCK_RAW
BAT_BUCK_RAW
A1
A2
BAT_36_P

R12
29.4k
BAT_CONFIG
BAT_P_RET

BAT_BOOT_R
C15
47n/25V
BAT_SW_BOOT

BAT_BUCK_RAW
C18
270u/63V
BAT_P_RET

BAT_FUSED_P
C19
100u/100V
BAT_P_RET

R13
100k
BAT_VDDA
BAT_COMP

R31
4.02k
BAT_FB_SERIES
BAT_FB

BAT_FUSED_P
C9
4.7u/100V
BAT_P_RET

BAT_FUSED_P
C10
4.7u/100V
BAT_P_RET

BAT_FUSED_P
C11
10n/100V
BAT_P_RET

BAT_FUSED_P
C12
100n/100V
BAT_P_RET

BAT_VDDA
C14
100n/100V
BAT_P_RET

BAT_VCC
C13
4.7u/16V
BAT_P_RET

BAT_BUCK_RAW
C16
4.7u/100V
BAT_P_RET

BAT_BUCK_RAW
C17
4.7u/100V
BAT_P_RET

BAT_36_P
C20
270u/63V
BAT_P_RET

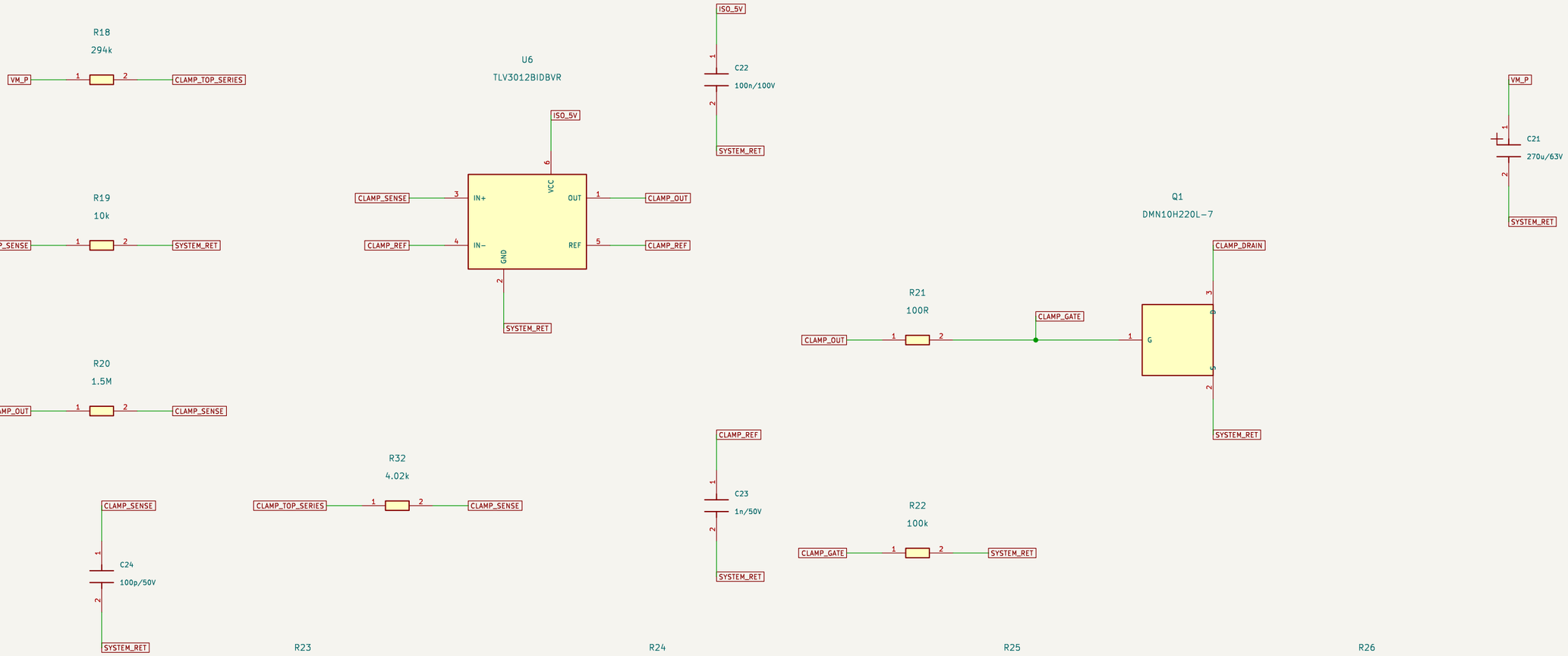
SW4 is internally connected to SW1/2/3 in U5. BAT_SW_BOOT keeps the bootstrap return local; do not route it around the inductor.

36.4816 V before D7; approximately 36 V at VM under load. No output boost below dropout.
6 A converter design target; 2.54mm headers limit the path to at most 3 A; jumper leads may be lower. Initial bring-up: <=2 A.
AUTO mode / internal compensation. Loop stability, cooling and load transients require bench verification.

BAT_FUSED_P
BAT_BUCK_RAW

84 x 128 mm | 4 layers / 1 oz | 36 V nominal | isolated low-voltage system
NIU intermediate board
Sheet: /Buck 36V/
File: Buck_36V.kicad_sch
Title: 36 V nominal / 6 A converter design target
Size: A3 | Date: 2026-10-08 | **Rev: 0 PROTOTYPE**
KiCad E.D.A. 9.0.7 | Id: 5/7

06 / VM TRANSIENT BUFFER (NOT A BRAKING SYSTEM)



Nominal turn−on about 38.50 V; turn−off about 37.51 V. 8 x 1k = 125 ohm dump bank.
Approximately 0.31 A / 12 W maximum instantaneous sink near threshold; not a regen brake.
Use only for brief transients until pulse−energy and thermal tests pass. It cannot absorb sustained backdrive.
Powered from always−on isolated 5 V. Motor PCB relay voltage/current limits remain unchanged.

84 x 128 mm | 4 layers / 1 oz | 36 V nominal | isolated low−voltage system

NIU intermediate board

Sheet: /VM Transient Clamp/
File: VM_Transient_Clamp.kicad_sch

Title: VM transient buffer and limited−energy shunt

Size: A3	Date: 2026−10−08	**Rev: 0 PROTOTYPE**
KiCad E.D.A. 9.0.7		Id: 6/7

07 / BATTERY DRAW + CHARGING CURRENT

R33
5m / 2W / 1%
BAT_SENSE_SOURCE
1
2
BAT_FUSED_P

R34
5m / 2W / 1%
CHG_SENSE_SOURCE
1
2
CHARGER_P

R35
100R
BAT_SENSE_SOURCE
1
2
BAT_INP_F

U7
INA237 / 0x40
BAT_INP_F
10
IN+
A1
1
BAT_P_RET
BAT_INM_F
9
IN−
A0
2
BAT_P_RET
BAT_VBUS_F
8
VBUS
ALERT
3
BAT_SENSE_3V3
6
VS
SDA
4
BAT_SDA
BAT_P_RET
7
GND
SCL
5
BAT_SCL

R41
4.02k
BAT_SENSE_3V3
1
2
BAT_SDA

R36
100R
BAT_FUSED_P
1
2
BAT_INM_F

R42
4.02k
BAT_SENSE_3V3
1
2
BAT_SCL

R37
100R
CHG_SENSE_SOURCE
1
2
CHG_INP_F

U8
INA237 / 0x41
CHG_INP_F
10
IN+
A1
1
SYSTEM_RET
CHG_INM_F
9
IN−
A0
2
SENSE_3V3
CHG_VBUS_F
8
VBUS
ALERT
3
SENSE_3V3
6
VS
SDA
4
SOC_SDA
SYSTEM_RET
7
GND
SCL
5
SOC_SCL

R43
4.02k
SENSE_3V3
1
2
SOC_SDA

R38
100R
CHARGER_P
1
2
CHG_INM_F

R44
4.02k
SENSE_3V3
1
2
SOC_SCL

R39
100R
BAT_FUSED_P
1
2
BAT_VBUS_F

C32
10n / 50V
BAT_INP_F
1
2
BAT_INM_F

C34
100n / 100V
BAT_VBUS_F
1
2
BAT_P_RET

C25
100n / 100V
BAT_SENSE_3V3
1
2
BAT_P_RET

R40
100R
CHARGER_P
1
2
CHG_VBUS_F

C33
10n / 50V
CHG_INP_F
1
2
CHG_INM_F

C35
100n / 100V
CHG_VBUS_F
1
2
SYSTEM_RET

C26
100n / 100V
SENSE_3V3
1
2
SYSTEM_RET

C27
100n / 100V
BAT_SENSE_3V3
1
2
BAT_P_RET

U9
ISO1640B-Q1
BAT_SENSE_3V3
1
VCC1
VCC2
8
SENSE_3V3
BAT_SDA
2
SDA1
SDA2
7
SOC_SDA
BAT_SCL
3
SCL1
SCL2
6
SOC_SCL
BAT_P_RET
4
GND1
GND2
5
SYSTEM_RET

C28
100n / 100V
SENSE_3V3
1
2
SYSTEM_RET

J12
SOC SDA / SCL
1
2
SOC_SDA
SOC_SCL

U11
TLV75533 / LOCAL 3V3
ISO_5V
1
IN
OUT
5
SENSE_3V3
ISO_5V
3
EN
SYSTEM_RET
2
GND
NC
4

C29
4.7u / 100V
AUX_IN_P
1
2
BAT_P_RET

U10
TPS7A4001 / 3.31V
AUX_IN_P
8
IN
OUT
1
BAT_SENSE_3V3
AUX_IN_P
5
EN
FB
2
BAT_SENSE_FB
BAT_P_RET
4
GND
NC
3
BAT_P_RET
BAT_P_RET
9
PAD
NC
6
BAT_P_RET
NC
7
BAT_P_RET

R45
18.2k / 1%
BAT_SENSE_3V3
1
2
BAT_SENSE_FB

C36
10u / 16V
ISO_5V
1
2
SYSTEM_RET

C37
10u / 16V
SENSE_3V3
1
2
SYSTEM_RET

C30
10u / 16V
BAT_SENSE_3V3
1
2
BAT_P_RET

C31
10u / 16V
BAT_SENSE_3V3
1
2
BAT_P_RET

R46
10k
BAT_SENSE_FB
1
2
BAT_P_RET

U7: 0x40 battery side; U8: 0x41 charger side. Calibrate both 5 mOhm shunts.
U11 supplies system-side 3.3 V from isolated 5 V; J13 removed.
U10 remains the separate battery-side supply. Never bridge the returns.
Charger voltage is not cell-stack voltage when the charge relay is open.

NIU intermediate board

Sheet: /SOC Sensing/
File: SOC_Sensing.kicad_sch

Title: Isolated battery sensing / charger current / I2C

Size: A3	Date: 2026-10-08	**Rev: 0 PROTOTYPE**
KiCad E.D.A. 9.0.7		Id: 7/7